

STRUCTURAL OBSTRUCTIONS TO STANDARD-MODEL DERIVATION FROM BINARY ICOSAHEDRAL ORBIFOLDS AND MODULAR-FLAVOR EXTENSIONS

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ABSTRACT. We exhaustively test the class of candidate Standard Model (SM) derivations based on E_8 gauge theory compactified on $M^4 \times S^3/2I \times T^2(\tau=i)$, where $2I = \mathrm{SL}(2, \mathbb{F}_5)$ is the binary icosahedral group, including all natural extensions to modular-flavor symmetries, alternative compactification paradigms, eclectic non-split groups, twisted modular tensor categories, and 2-group constructions. In 80 distinct rigorous computational investigations across 27 rounds (as of June 2026), we establish three headline structural impossibility theorems, each closing the program from an independent mathematical direction. *Theorem A* (Arithmetic Obstruction): characters of finite-dimensional representations of any finite group G live in the cyclotomic field $\mathbb{Q}(\zeta_{\exp(G)})$; for $2I$, $\exp(2I) = 60$ with prime support $\{2, 3, 5\}$, while $\sin(\pi/14) \in \mathbb{Q}(\zeta_{14})$ requires the prime 7, and by Goursat’s lemma and the Bantay–Coste–Gannon–Ruelle character cap this prime separation persists through all known finite-symmetry constructions. *Theorem B* (Frobenius–Schur chirality obstruction): every faithful irreducible representation of $2I$ is quaternionic and no irreducible is of complex type, so the equivariant Dirac index of any $2I$ worldsheet orbifold vanishes identically, independently of level matching, gauge embedding, and modular invariance. *Theorem C* (centralizer obstruction): $F_4 \times G_2$ is a maximal regular dual pair in E_8 with trivial centralizer, so no heterotic bundle can realize it as the visible 4D gauge group. Five supporting propositions — surviving-algebra structure, generation–gauge incompatibility, geometry-independence, CRT factorization, and the modular-tower plateau — complete the closure. The contributions are negative: a class of candidate theories ruled out, with the algebraic constraints any successful future program must satisfy made explicit.

1. INTRODUCTION

The Standard Model has nineteen empirically determined parameters whose values exhibit suggestive numerical regularities — among them the Cabibbo angle $\varepsilon \approx \sin(\pi/14)$ (1.1% match), the PMNS solar mixing $\sin^2 \theta_{12} \approx 3/\pi^2$ (0.003 σ match), the Koide lepton-mass relation $Q \approx 2/3$ (with the angle parameter $\theta_K \approx 2/9$ to 0.0033%), and the digital-root density $3/8$ of Pisano periods of Fibonacci sequences modulo prime levels.

A natural program of work attempts to derive these regularities from compactification of E_8 gauge theory on $M^4 \times S^3/2I \times T^2(\tau=i)$, where $2I = \mathrm{SL}(2, \mathbb{F}_5)$ is the binary icosahedral group of order 120. The program leverages the McKay correspondence $2I \leftrightarrow E_8$ [14] and the spectral gap on the Poincaré homology sphere $S^3/2I$ [12] to motivate phenomenological constructions producing Yukawa structures, Coleman–Weinberg potentials, and threshold corrections.

This paper presents the result of an exhaustive computational stress test of this class of constructions, including all natural extensions:

- Alternative binary polyhedral orbifolds ($2T \rightarrow E_6$, $2O \rightarrow E_7$);
- Alternative compactification paradigms (Calabi–Yau, F-theory, G_2 , heterotic bundle, asymmetric orbifold);
- 4D discrete flavor symmetry reformulations ($2I$, A_5);
- Alternative discrete groups ($\mathrm{PSL}(2, 7)$, Klein quartic, $(2, 3, 7)$ triangle, E_{10} hyperbolic, Mathieu sporadics);
- Triality-invariant orbifolds via the $\mathbb{Z}_3$ outer automorphism of D_4 ;
- ML-accelerated exhaustive scan over 4,688 configuration parameters;
- Modular flavor symmetry at levels $\Gamma(N)$ for $N \in \{5, 7, 8, 10, 14, 20, 35, 70, 140\}$;

- The fusion group $\Gamma_{70} = \mathrm{SL}(2, \mathbb{Z}/70) \cong S_3 \times 2I \times \mathrm{SL}(2, 7)$;
- Moonshine constraint via $2I \rightarrow 2.A_5 \rightarrow 2.\mathrm{Co}_1 \rightarrow \mathbb{M}$ (the Monster);
- Eclectic flavor groups $(\Delta(54), \Delta(108), \Delta(150))$;
- Twisted Drinfeld centers $\mathcal{Z}(\mathrm{Vec}_{2I}^\omega)$ for all $\omega \in H^3(2I, U(1)) = \mathbb{Z}/120$;
- Metaplectic covers $\mathrm{Mp}(2, \mathbb{Z}/14)$;
- 2-group constructions with $\pi_1, \pi_2 \in \{2I, \mathrm{SL}(2, 7)\}$.

Across 80 distinct rigorous computational investigations spanning 27 rounds (as of June 2026) and all of these directions, three headline structural impossibility theorems characterize the obstruction, each closing the program from an independent mathematical direction: the *Arithmetic Obstruction* (Theorem A), which bars any finite-symmetry construction from fusing icosahedral character data with the cyclotomic-14 home of $\sin(\pi/14)$; the *Frobenius–Schur chirality obstruction* (Theorem B), which closes all worldsheet realizations of $2I$, whatever level-matching, gauge-embedding, and modular-invariance choices are made; and the *centralizer obstruction* (Theorem C), which closes heterotic bundle realization of the $F_4 \times G_2$ folded gauge group. Five supporting propositions — the surviving-algebra scan, the generation–gauge incompatibility, the geometry-independent obstruction, the CRT factorization of square-free modular flavor groups, and the modular-tower plateau — complete the closure and are collected in Section 4.

2. NOTATION AND PRELIMINARIES

Throughout we work with finite groups G , their representation theory over $\mathbb{C}$, and associated cyclotomic fields. We write $\exp(G)$ for the exponent of G (the smallest n such that $g^n = e$ for all $g \in G$), and $\zeta_n = e^{2\pi i/n}$.

For $G = 2I$, $\exp(2I) = 60$.

Definition 2.1 (Frobenius–Schur indicator). For a finite group G and irreducible complex character χ of G , the *Frobenius–Schur indicator* is

$$\nu(\chi) = \frac{1}{|G|} \sum_{g \in G} \chi(g^2) \in \{+1, 0, -1\}.$$

One has $\nu(\chi) = +1$ iff χ is afforded by a representation of *real type* (admitting a G -invariant non-degenerate symmetric bilinear form), $\nu(\chi) = -1$ iff it is of *quaternionic type* (G -invariant skew form), and $\nu(\chi) = 0$ iff it is of *complex type* ($\chi \neq \bar{\chi}$) [6, 11].

We rely on the following standard results:

- *McKay correspondence* [14]: the 9 irreducible representations of $2I$ correspond bijectively to the 9 nodes of the affine E_8 Dynkin diagram, with tensor product structure mirroring the affine Cartan matrix.
- *Klein invariant ring on $S^3/2I$* [12]: the invariant ring of polynomial functions in two complex variables under $2I \subset \mathrm{SU}(2)$ is $\mathrm{Sym}^\bullet(\mathbb{C}^2)^{2I} = \mathbb{C}[I_{12}, I_{20}, I_{30}]/(R_{60})$, giving the Molien series

$$M_{2I}(t) = \frac{1 + t^{30}}{(1 - t^{12})(1 - t^{20})}.$$

- *D_4 triality*: the unique $\mathbb{Z}_3$ outer automorphism of any simple Dynkin diagram (in fact, of any simple complex Lie algebra), acting on D_4 by cyclically permuting the three 8-dimensional representations $8_v, 8_s, 8_c$.

3. THREE STRUCTURAL IMPOSSIBILITY THEOREMS

This section states and proves the three headline structural impossibility theorems. Each closes the program from an independent mathematical direction: arithmetic (character fields), representation theory (chirality indices), and Lie theory (centralizers in E_8).

3.1. Theorem A: the arithmetic obstruction.

Theorem A (Arithmetic Obstruction). *Let G be any finite group, and let χ be a character of any finite-dimensional complex representation of G . Then all values of χ lie in the cyclotomic field $\mathbb{Q}(\zeta_{\exp(G)})$.*

If we further demand that $\chi(g) = \alpha$ for some specific algebraic number $\alpha \in \mathbb{Q}(\zeta_n)$ requiring a prime p that does not divide $\exp(G)$, then no such G can have α as a character value of any irreducible representation.

In particular, no irreducible representation of any finite group G obtained from 2I by:

- (a) *non-split central extension,*
- (b) *eclectic semidirect product $N \rtimes Q$,*
- (c) *metaplectic cover $\text{Mp}(2, \mathbb{Z}/N)$,*
- (d) *twisted Drinfeld center $\mathcal{Z}(\text{Vec}_G^\omega)$ for any $\omega \in H^3(G, U(1))$, or*
- (e) *categorical 2-group with $\pi_1, \pi_2 \in \{2I, \text{SL}(2, 7)\}$*

admits a single irreducible character with values in both $\mathbb{Q}(\zeta_5)$ (containing φ) and $\mathbb{Q}(\zeta_7)$ (containing $\sin(\pi/14)$).

Proof. The first claim is elementary: for any representation ρ of G and any $g \in G$, $\rho(g)^{\exp(G)} = I$, so every eigenvalue of $\rho(g)$ is an $\exp(G)$ -th root of unity, and the character value $\chi(g) = \text{Tr} \rho(g)$ is a sum of such roots of unity, hence lies in $\mathbb{Q}(\zeta_{\exp(G)})$. The Bantay–Coste–Gannon–Ruelle character cap [3, 7] extends this to twisted Drinfeld centers: characters of $\mathcal{Z}(\text{Vec}_G^\omega)$ lie in $\mathbb{Q}(\zeta_{\exp(G)})$ regardless of the cocycle ω .

For (a), non-split central extensions $\tilde{G} \rightarrow G$ with kernel A have $\exp(\tilde{G}) \mid \exp(G) \cdot \exp(A)$. Since 2I has trivial Schur multiplier $H^2(2I, U(1)) = 0$, no central extension of 2I exists with non-trivial kernel coupled non-trivially to the 2I part, and the prime support is unchanged.

For (b), semidirect products $N \rtimes Q$ with action $\alpha : Q \rightarrow \text{Aut}(N)$ have $\exp(N \rtimes Q) \mid \exp(N) \cdot \exp(Q)$; the prime support of $N \rtimes Q$ is the union of those of N and Q . To include both prime support $\{2, 3, 5\}$ (icosahedral) and $\{7\}$ (cyclotomic-14), the group G' must factor as a product of factors covering each prime. By Goursat’s lemma, irreducible representations of such a product factor as outer tensor products, so no single irrep carries cross-prime data in its character.

For (c), metaplectic covers $\text{Mp}(2, \mathbb{Z}/N)$ have a non-trivial cocycle concentrated at $p = 2$ (the Weil representation). For $N = 14$, the cocycle decouples the $p = 2$ sector from the $p = 7$ sector, so $\text{Mp}(2, \mathbb{Z}/14) \cong \text{Mp}(2, \mathbb{Z}/2) \times \text{SL}(2, \mathbb{Z}/7) \cong D_6 \times \text{SL}(2, 7)$ as a pure CRT product.

For (d), the Bantay–Coste–Gannon–Ruelle theorem gives the character cap.

For (e), 2-groups with $\pi_1 = 2I$ and $\pi_2 = \text{SL}(2, 7)$ have Postnikov data $H^3(\pi_1, \pi_2) = H^3(2I, \text{SL}(2, 7))$. With π_2 abelianized $\text{SL}(2, 7)^{ab} = 0$ (perfect group), the 3-cocycle is trivial; with the non-abelian π_2 , the action $2I \rightarrow \text{Out}(\text{SL}(2, 7))$ factors through $2I^{ab} = 0$, again trivial. Either way the 2-group reduces to a direct product, governed by Proposition 4.7. $\square$

Corollary. *The class of finite-symmetry constructions admitting both icosahedral content and the cyclotomic-14 home of $\sin(\pi/14)$ as character values of a single irreducible representation is empty. Any genuine fusion requires either:*

- *infinite-dimensional structures (vertex operator algebras, modular forms with level-mixing operators, Maass-form lifts);*
- *multiple vacua at distinct moduli points (string landscape); or*
- *acceptance that the empirical numerical coincidence has no finite-symmetry-algebraic origin.*

Remark 3.1. The obstruction can be sharpened arithmetically: $\sin(\pi/14)$ is not even an algebraic integer (its minimal polynomial $8x^3 - 4x^2 - 4x + 1$ is not monic), while every character value of every finite group is a sum of roots of unity and hence an algebraic integer. So $\sin(\pi/14)$ can never appear as a character value of *any* finite group whatsoever; the cyclotomic argument of Theorem A additionally shows that no 2I-derived character value can generate $\mathbb{Q}(\zeta_{14})$ (verified numerically in the supplementary script `supplement_verify.py`).

TABLE 1. Frobenius–Schur data for the nine irreducible representations of $2I = \mathrm{SL}(2, 5)$ (z denotes the central involution -1). Real type ($\nu = +1$) coincides with factoring through $A_5 = 2I/\{\pm 1\}$; quaternionic type ($\nu = -1$) coincides with faithfulness.

Irrep	$\chi(1)$	$\chi(z)$	Factors through A_5	$\nu(\chi)$
1	1	+1	yes	+1
3	3	+3	yes	+1
3'	3	+3	yes	+1
4	4	+4	yes	+1
5	5	+5	yes	+1
2	2	-2	no (faithful)	-1
2'	2	-2	no (faithful)	-1
4'	4	-4	no (faithful)	-1
6	6	-6	no (faithful)	-1

3.2. Theorem B: the Frobenius–Schur chirality obstruction. Theorem A closes the arithmetic route, and the supporting propositions of Section 4 close the geometric compactification routes. The remaining escape — realizing $2I$ *non-geometrically* as an exact worldsheet symmetry — requires a separate representation-theoretic closure. The obstruction is the Frobenius–Schur type (Definition 2.1) of the irreducible representations of $2I$.

Theorem B (Frobenius–Schur chirality obstruction). *Every irreducible complex representation of $2I$ is of real or quaternionic type; none is of complex type. Equivalently, the Frobenius–Schur indicators of the nine irreducible characters (Table 1) are $\nu = +1$ for the five irreps of dimensions 1, 3, 3', 4, 5 — exactly those factoring through $A_5 = 2I/\{\pm 1\}$ — and $\nu = -1$ for the four faithful irreps of dimensions 2, 2', 4', 6; in particular, every faithful irrep of $2I$ is quaternionic. Consequently, for any worldsheet construction carrying a $2I$ action — Gepner minimal model, free-fermionic, direct $2I$ orbifold of T^d , or McKay/ALE lift — the equivariant index of the Dirac operator in any $2I$ -sector of real or quaternionic type vanishes identically, and no such construction delivers chiral 4D fermions through the $2I$ action, independently of level matching, gauge embedding, and modular invariance.*

Proof. We first record the kernel structure of $2I$. The normal subgroups of $2I = \mathrm{SL}(2, 5)$ are exactly 1, the center $Z = \{\pm 1\}$, and $2I$ itself, since $2I/Z \cong A_5$ is simple and $|Z| = 2$. An irreducible representation ρ with $\chi_\rho(z) = \chi_\rho(1)$ contains Z in its kernel and factors through A_5 ; one with $\chi_\rho(z) = -\chi_\rho(1)$ (z acts as -1) is faithful. Direct evaluation of

$$\nu(\chi_\alpha) = \frac{1}{120} \sum_{i=1}^9 |C_i| \chi_\alpha(g_i^2),$$

where $C_1, \dots, C_9$ are the conjugacy classes of sizes 1, 1, 12, 12, 12, 12, 20, 20, 30 with representatives g_i , on the character table of $2I$ (computed from the class algebra of the 120 unit quaternions; supplementary script `supplement_verify.py`; consistent with [6]) gives Table 1. Two independent consistency checks confirm the assignment. (i) The Frobenius count $\sum_\alpha \nu(\chi_\alpha) \chi_\alpha(1) = \#\{g \in 2I : g^2 = 1\} = 2$ (the identity and z), which is satisfied by the assignment of Table 1 and violated by any other choice of signs. (ii) The five irreps of type $\nu = +1$ are exactly those with $\chi(z) = +\chi(1)$, i.e. the pullbacks of the five irreps of A_5 along $\pi : 2I \rightarrow A_5$; for these, $\nu_{2I}(\chi \circ \pi) = \nu_{A_5}(\chi)$ because the projection is two-to-one (both the sum and $|G|$ double), and every A_5 irrep is real: $\sum_{\text{irreps}} d = 1 + 3 + 3 + 4 + 5 = 16 = \#\{a \in A_5 : a^2 = 1\}$ forces $\nu = +1$ on each by the same Frobenius count. Hence $\nu \neq 0$ for all nine irreps: $2I$ has no complex-type irrep, and by the kernel dichotomy the faithful irreps are exactly the quaternionic ones.

For the index claim, let D be the orbifold Dirac operator and let the fermions transform in a $2I$ -representation V of real or quaternionic type: V carries an antilinear $2I$ -equivariant involution

J with $J^2 = +1$ (real) or $J^2 = -1$ (quaternionic). By the equivariant Atiyah–Singer index theorem, the net chirality is the equivariant index

$$\text{ind}_{2I}(D_V) = \frac{1}{|2I|} \sum_{g \in 2I} \chi_V(g) \text{tr}_g(\text{twist factors}),$$

a virtual character of $2I$. The twist factors enter through real data (the transverse rotation representation of the orbifold), and χ_V is real-valued since $\nu(\chi_V) = \pm 1$. Complex conjugation therefore acts as the identity (real case) or as an antilinear sign structure (quaternionic case) on V , and the chirality projection commutes with this involution, forcing every left-chiral zero mode to pair with a right-chiral conjugate: the index is paired with its conjugate and vanishes identically. This is the same mechanism by which gauge groups acting through real representations (e.g. F_4) cannot generate chirality without an external twist [17, 21].

Universality is now immediate: the index argument uses only the Frobenius–Schur type of the $2I$ action — not the moduli of level-matching shifts, gauge embeddings, or discrete torsion — so any choice of these auxiliary data leaves the equivariant index zero. The per-route secondary obstructions are recorded in Remark 3.2. $\square$

Remark 3.2 (Per-route secondary obstructions). Theorem B is universal, but each standard worldsheet route also fails on its own independent grounds:

- (1) *Gepner / $\mathcal{N} = 2$ minimal models*: the published catalogue realizes A_5 (not $2I$) via permutation orbifolds of $(k)^5$; A_5 does not lift to its binary double cover by permutation, and a direct $2I$ action through the 5-dimensional irrep violates the $\mathcal{N} = 2$ fusion rules [9, 13].
- (2) *Free fermions (NAHE basis)*: twist vectors must lie in $(1/N)\mathbb{Z}$ -lattices, and the Faraggi catalogue realizes only S_3, S_4, A_4, D_n — never $2I$ [8]. The McKay $j = 3/2$ embedding is 8-dimensional *real*, so the fermion-number index vanishes directly.
- (3) *Direct $2I$ orbifold of T^d* : level matching requires $h_g = 9/100$ in an order-5 twisted sector for $d = 4$ (a precisely tuned gauge twist), and $d = 6$ requires a $2I$ -invariant CY_3 , which does not exist (Proposition 4.5).
- (4) *McKay/ALE lift*: heterotic ALE compactification enhances to E_8 at $2I$ orbifold points [17], but K3 has $\text{SU}(2)$ holonomy — $\mathcal{N} = 2$ in 4D, hence non-chiral — and the $2I$ enhancement occurs only at isolated fixed points (Mukai [15]), never as a global worldsheet symmetry.

Corollary. *No $2I$ worldsheet orbifold — geometric or non-geometric — can deliver chiral 4D fermions through the $2I$ action. Combined with Proposition 4.5 and the Mukai–Nikulin closure [15, 16] of the geometric route, the icosahedral content of the framework survives only as a lattice/algebra symmetry (icosian E_8 lattice, Yukawa coefficient ring), never as a dynamical worldsheet or manifold symmetry; any three-generation realization must source its chirality from an unrelated mechanism (e.g. abelian Wilson lines on a $\mathbb{Z}_3$ -quotient CY_3 with $\chi = -6$).*

3.3. Theorem C: the $F_4 \times G_2$ centralizer obstruction. The last dynamical route is heterotic: realize the folded gauge group $F_4 \times G_2 \subset E_8$ as the visible 4D gauge group via a stable holomorphic bundle on a three-generation ($\chi = -6$) Calabi–Yau threefold. It is closed by a centralizer argument.

Theorem C (Centralizer obstruction). *Let X be any Calabi–Yau threefold and $V \rightarrow X$ a stable holomorphic vector bundle with structure group $H \subset E_8$, so that (in the standard heterotic dictionary [22]) the visible 4D gauge group is $G_{4D} = C_{E_8}(H)$. Since $F_4 \times G_2$ is a maximal regular dual pair in E_8 [18],*

$$C_{E_8}(F_4 \times G_2) = \{e\},$$

the inclusion $F_4 \times G_2 \subseteq G_{4D}$ forces $H = \{e\}$: no bundle, no symmetry breaking, full E_8 unbroken in 4D — which defeats the purpose of the compactification. Moreover, the Chaudhuri–Hockney–Lykken alternative [5] — an asymmetric $\mathbb{Z}_2$ outer-automorphism orbifold that does realize F_4 dynamically from $E_8 \times E_8$ — requires a freely acting $\mathbb{Z}_2$ isometry of X ; among three-generation candidates, the Tian–Yau manifold M_{TY} [19] admits only its defining freely acting $\mathbb{Z}_3$ [10], no $\chi = -6$ threefold with a free $\mathbb{Z}_2$ appears in the Anderson–Gray–Lukas–Palti catalogue [2], and $\mathbb{Z}_2$

quotients of $\chi = -12$ parents fail the three-generation index count. Hence no construction in this class realizes $F_4 \times G_2$ as a dynamical visible 4D gauge group.

Proof. The decomposition

$$248 = (52, 1) \oplus (1, 14) \oplus (26, 7)$$

($52 + 14 + 182 = 248$) realizes $F_4 \times G_2 \subset E_8$ as a symmetric pair in which each factor is the full centralizer of the other (a maximal regular dual pair; Slansky [18], Table 14). Hence

$$C_{E_8}(F_4 \times G_2) = C_{E_8}(F_4) \cap C_{E_8}(G_2) = \mathfrak{g}_2 \cap \mathfrak{f}_4 = \{0\}$$

at the level of Lie algebras, so the centralizer group is trivial. Under the Witten dictionary [22], the unbroken gauge group of a heterotic compactification with bundle structure group $H \subset E_8$ is the centralizer $C_{E_8}(H)$: the covariant derivative of V must commute with the surviving gauge fields. Thus $F_4 \times G_2 \subseteq C_{E_8}(H)$ holds if and only if $H \subseteq C_{E_8}(F_4 \times G_2) = \{e\}$, i.e. H is trivial and the visible group is all of E_8 . No non-trivial stable bundle on any Calabi–Yau threefold realizes $F_4 \times G_2$ as the visible 4D gauge group.

For the CHL alternative: the asymmetric $\mathbb{Z}_2$ twist by the outer automorphism of $E_8 \times E_8$ (factor exchange combined with a lattice shift) realizes F_4 dynamically [5, 17], but consistency of the asymmetric orbifold requires the geometric $\mathbb{Z}_2$ to act freely on X . The Tian–Yau manifold M_{TY} (CICY-7884/ $\mathbb{Z}_3$; $\chi = -6$, $h^{1,1} = 6$, $h^{2,1} = 9$, $\pi_1 = \mathbb{Z}_3$) [19] has its freely acting symmetries exhausted by the defining $\mathbb{Z}_3$ quotient [10]: no free $\mathbb{Z}_2$ exists. The Anderson–Gray–Lukas–Palti systematic search over complete intersection Calabi–Yau threefolds [2] contains no $\chi = -6$ candidate admitting a freely acting $\mathbb{Z}_2$. A freely acting $\mathbb{Z}_2$ on a $\chi = -12$ parent would restore $\chi = -6$ after the quotient, but the generations are then counted by the $\mathbb{Z}_2$ -equivariant (CHL-twisted) index, which is not half the untwisted index: the projection onto $\mathbb{Z}_2$ -invariant states does not reproduce three chiral generations. All alternatives fail. $\square$

Corollary. *The $F_4 \times G_2$ fold can survive only as a matter-content classification scheme within a conventional heterotic construction ($SU(5) / SO(10) / E_6$ bundle with abelian $\mathbb{Z}_3$ Wilson line on M_{TY}); the visible gauge group is then a conventional GUT, not $F_4 \times G_2$. The framework’s distinct dynamical content — the folded gauge group and the associated Yukawa reduction — is unrealizable; only the algebraic layer (icosian E_8 lattice, $\theta_{QCD} = 0$, Yukawa coefficient ring) survives, as detailed in the companion paper [20].*

4. SUPPORTING PROPOSITIONS

The following five results, previously framed as standalone theorems, support and sharpen the three headline obstructions: they close the surviving-algebra, generation-counting, geometric, CRT-factorization, and modular-tower escape routes that Theorems A–C leave ajar.

4.1. Wall 1: surviving algebra under $2I$ orbifolds.

Proposition 4.1 (Wall 1). *Let $\iota : 2I \rightarrow SU(2) \subset E_8$ be any embedding factoring through $SU(2)$, and let $\mathfrak{g}_{\text{surv}} = C_{E_8}(\iota(2I))$ denote the centralizer of the image of $2I$ in E_8 .*

For the natural $j = 3/2$ embedding (in which $2I$ acts on the 4-dimensional irreducible representation of $SU(2)$),

- *Within $D_8 \subset E_8$, $\mathfrak{g}_{\text{surv}} \cap D_8 = A_1 \oplus D_4$ (dimension 31, rank 5);*
- *The full E_8 centralizer is $F_4 \oplus \mathfrak{u}(1)^3$ (dimension 55, rank 7).*

For all 56 enumerated $SU(2)$ -embeddings into the maximal subalgebras of E_8 (the chains D_8 , $E_7 \times A_1$, $E_6 \times A_2$, A_8 , $D_5 \times A_3$, $A_4 \times A_4$, $F_4 \times G_2$), no $\mathfrak{g}_{\text{surv}}$ contains a simple factor with Coxeter number $h = 14$.

Proof. We construct $2I$ as 120 unit quaternions, embed in $SU(2)$ via the standard quaternionic representation, and compute the centralizer in $\mathfrak{su}(16) = D_8$ explicitly via the linear condition

$$\rho(g)X\rho(g)^T = X \quad \text{for all } g \in 2I, \quad X \in \mathfrak{su}(16).$$

For $j = 3/2$ the centralizer-within- D_8 has dimension 31 (basis verified by explicit kernel of the linear map $X \mapsto X - \rho(g)X\rho(g)^T$ over all g in conjugacy class representatives). Identification as

$A_1 \oplus D_4$ follows from rank 5 and matching the Cartan structure (3-dim A_1 plus 28-dim D_4 , total 31). The 24 additional generators in 128_s (half-spinor of D_8) that are $2I$ -invariant contribute to the centralizer-within- E_8 beyond D_8 , giving total dimension 55. Identifying as $F_4 \oplus \mathfrak{u}(1)^3$ at rank 7 follows from comparing dimensions and ranks against Slansky's tables [18].

For the 56-embedding scan: we enumerate the seven maximal-subalgebra chains of E_8 and within each consider the natural $SU(2)$ subgroups (principal, sub-principal, fundamental, regular extensions). For each candidate we compute $\mathfrak{g}_{\text{surv}}$ via centralizer computation. The Coxeter numbers h of the surviving simple factors take values in $\{0, 2, 3, 4, 6, 10, 12, 18, 30\}$. Notably, $h = 14$ never appears: the simple Lie algebras with $h = 14$ are $\{B_7, C_7, D_8, A_{13}\}$. Of these, D_8 and A_{13} have rank exceeding 7 (and so cannot embed in E_7 of rank 7), while B_7 and C_7 are not subalgebras of E_7 in Dynkin's classification of subalgebras. $\square$

Corollary 4.2. *The framework's claim $\varepsilon = \sin(\pi/h(D_8)) = \sin(\pi/14)$ fails: D_8 is not the surviving algebra in any non-trivial $2I$ orbifold of E_8 , and $h = 14$ is unreachable as a Coxeter number of any surviving simple factor.*

4.2. Generation–gauge incompatibility ($\mathbb{Z}_3$ tripling).

Proposition 4.3 (Generation–gauge incompatibility). *Let $G \subset SU(2)$ be any binary polyhedral group ($G \in \{2T, 2O, 2I\}$). For any embedding $\iota : G \rightarrow SU(2) \rightarrow E_n$ with E_n the McKay-correspondent exceptional Lie algebra ($E_n \in \{E_6, E_7, E_8\}$), the conditions*

- (i) *the surviving gauge algebra $\mathfrak{g}_{\text{surv}} = C_{E_n}(\iota(G))$ contains the Standard-Model gauge algebra $\mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$; and*
- (ii) *the abelianization G^{ab} acts non-trivially on the surviving matter (enabling a $\mathbb{Z}_3$ generation-tripling mechanism)*

are mutually exclusive.

Proof. For (ii) to hold, the action of G^{ab} on a representation V must factor through a non-trivial outer character. This requires the corresponding $SU(2)$ embedding into E_n to “stretch” V across multiple symmetric powers Sym^j of the defining $SU(2)$ representation. Among the binary polyhedrals, only $2T$ has non-trivial abelianization: $2T^{ab} = \mathbb{Z}_3$ (here $2T = \text{SL}(2, \mathbb{F}_3)$, with commutator subgroup Q_8 of order 8 and quotient $\mathbb{Z}_3$). The $\mathbb{Z}_3$ action on 27 of E_6 is exact $(3, 3, 3)$ only for the principal A_1 embedding (with characters $1, \omega, \omega^2$ on the three irreducible $\mathbb{Z}_3$ -eigenspaces); for any other embedding, the $\mathbb{Z}_3$ acts trivially on the surviving matter.

For (i) to hold, the centralizer must be non-abelian and contain $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. The principal A_1 embedding has minimal centralizer, of rank $\text{rank}(E_n) - 1$ with predominantly abelian directions. For $2T \rightarrow E_6$ principal, the centralizer is $\mathfrak{u}(1)^4$, which contains no non-abelian SM gauge piece.

For other $SU(2)$ -embeddings, larger centralizers exist (e.g. $A_5 = \mathfrak{su}(6)$ of dim 35 in $2T \rightarrow E_6$), but the G^{ab} action on the matter is inherited through tensor structure and reduces to trivial on each irreducible component.

The strict inverse relation between (i) and (ii) follows: large centralizer $\Rightarrow$ low- j $SU(2)$ embedding $\Rightarrow$ matter in low- j reps where G^{ab} acts trivially; non-trivial G^{ab} action on matter $\Rightarrow$ principal $A_1 \Rightarrow$ centralizer is mostly abelian. $\square$

Corollary 4.4. *The “three generations from G^{ab} tripling” argument cannot be combined with SM gauge structure in any binary polyhedral orbifold of an exceptional Lie algebra.*

4.3. Geometry-independent obstruction.

Proposition 4.5 (Geometry-independent obstruction). *The binary icosahedral group $2I$ has two intrinsic properties that obstruct SM derivation in every compactification paradigm tested:*

- (a) *$2I \hookrightarrow SU(2)$ faithfully (i.e., $2I$ is a subgroup of $SU(2)$ without a non-trivial quotient through which it factors);*
- (b) *$2I$ is a perfect group: $2I^{ab} = 0$.*

Property (a) implies that any geometric action on a gauge fiber routes through an $SU(2)$ embedding, and the centralizer in any exceptional E_n generically kills chirality (the surviving matter

is real or pseudoreal, not complex). Property (b) implies trivial Schur multiplier $H^2(2I, U(1)) = 0$ and trivial first cohomology $\text{Hom}(2I, U(1)) = 0$, ruling out non-split central extensions, abelian Wilson lines, and spin-lift ambiguities.

Combined, these properties make every of the following compactification paradigms fail:

- (1) Calabi–Yau 3-fold with A_5 action;
- (2) F-theory with $2I$ on singular base locus;
- (3) M-theory on G_2 with $2I$ codim-4 singularity ([1]);
- (4) Heterotic with $2I$ vector bundle;
- (5) Asymmetric $T^6/2I$ orbifold.

Proof outline. For (1), Braun’s classification [4] of free A_5 actions on Calabi–Yau 3-folds shows none exist; non-free actions reinstate the centralizer obstruction of Proposition 4.1. For (2), the $2I$ action on a 7-brane locus routes through $\text{SU}(2) \subset E_8$ and inherits the centralizer failure. For (3), the codim-4 $2I$ singularity is McKay-natural but chirality in G_2 -holonomy compactifications comes from independent codim-7 data not constrained by $2I$. For (4), the heterotic bundle’s fundamental group requirement $\pi_1(X) \twoheadrightarrow 2I$ is not satisfied by any known Calabi–Yau 3-fold; even if such an X existed, the Atiyah–Singer index theorem applied to the pair (defining $\text{SU}(2)$ -rep, **56** of E_7) gives 0 chiral generations, since $\mathbf{2} \otimes \mathbf{2}$ contains no $2I$ -invariant subspace. For (5), all three escape hatches (Schur multiplier, abelianization, π_1 of T^6) vanish. $\square$

Corollary 4.6. *The framework’s geometric content cannot rescue SM derivation through any change of compactification paradigm.*

4.4. CRT factorization of square-free modular flavor groups.

Proposition 4.7 (CRT factorization of modular flavor). *Let $N = p_1 p_2 \cdots p_k$ be a square-free positive integer with distinct primes p_i . Then by the Chinese Remainder Theorem,*

$$\text{SL}(2, \mathbb{Z}/N) \cong \prod_{i=1}^k \text{SL}(2, \mathbb{Z}/p_i),$$

and every irreducible representation of $\text{SL}(2, \mathbb{Z}/N)$ is an outer tensor product of irreducible representations of the factors. In particular, no irreducible representation of $\text{SL}(2, \mathbb{Z}/N)$ has character values simultaneously involving distinct cyclotomic primes; cross-prime “fusion” of algebraic numbers (e.g. $\varphi \in \mathbb{Q}(\zeta_5)$ and $\sin(\pi/14) \in \mathbb{Q}(\zeta_7)$) cannot occur within any single irreducible component.

Proof. The CRT decomposition for $\text{SL}(2, \mathbb{Z}/N)$ over square-free N is standard. Irreducibility of outer tensor products follows from the Frobenius–Schur orthogonality relations: for irreducible representations ρ of A and σ of B , the character of $\rho \boxtimes \sigma$ is $\chi_{\rho \boxtimes \sigma}(a, b) = \chi_\rho(a) \chi_\sigma(b)$, and orthogonality gives that this is irreducible as a representation of $A \times B$ iff ρ and σ are irreducible. Algebraic factorization: the character ring of $A \times B$ is the tensor product of factor character rings, with characters living in the compositum $\mathbb{Q}(\zeta_{\exp(A)}) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_{\exp(B)}) = \mathbb{Q}(\zeta_{\text{lcm}(\exp(A), \exp(B))})$. For $\gcd(p_i, p_j) = 1$ (square-free N), this compositum has no cross-prime mixing in any single character value of an irreducible component. $\square$

Corollary 4.8. *The level-14 modular flavor group $\Gamma_{14} = S_3 \times \text{SL}(2, 7)$, the level-70 fusion group $\Gamma_{70} = S_3 \times 2I \times \text{SL}(2, 7)$, and all CRT-square-free modular flavor groups cannot fuse φ and $\sin(\pi/14)$ into a single irreducible representation. The naive “larger level $\Rightarrow$ greater fusion” rescue does not work in the square-free case.*

4.5. Modular tower plateau.

Proposition 4.9 (Empirical plateau). *Let $\chi_{\min}^2(N)$ be the simultaneous χ^2 -fit quality of a generic weight-2 modular flavor model at level $\Gamma(N)$, with 30 real $O(1)$ coefficients in addition to the modular parameter τ , fitted to the 14 SM observables (quark and lepton mass ratios, CKM and PMNS mixing angles, Δm^2 ratio, δ_{CP}). Empirically tested for $N \in \{5, 7, 8, 10, 14, 20, 35, 70, 140\}$,*

$$8.97 \times 10^5 \leq \chi_{\min}^2(N) \leq 2.21 \times 10^6,$$

i.e., $\chi_{\min}^2(N)$ plateaus across the entire range, with maximum/minimum ratio 2.5, despite an $\sim 7000\times$ growth in $\dim M_2(\Gamma(N))$ (from 22 at $N = 5$ to 168 192 at $N = 140$).

Remark 4.10. The structural reason: in any modular flavor model with k real $O(1)$ coefficients, increasing the level N adds modular forms whose contributions are linear in the $O(1)$ coefficients (after τ fixing). The fit χ^2 is a quadratic function of these coefficients with at most $k - r$ free directions absorbed by fitting (where r is the number of fixed structural relations from the Γ_N representation theory). For $k = 30$ and $r = O(1)$, the χ^2 minimum is dominated by the constant term, independent of $\dim M_2$.

Corollary 4.11. *The “go to higher level” rescue of modular flavor models does not constrain SM observables beyond the freedom of order-30 free coefficients. Without additional non-modular constraints (e.g. generalized CP, eclectic flavor, gauge-flavor unification), modular structure richness alone is absorbed by the parameter space.*

5. NUMERICAL COINCIDENCES: STATUS AND HONEST FRAMING

The original framework was motivated by four numerical matches in SM data:

Coincidence	Match level	Status after this work
$\varepsilon \approx \sin(\pi/14)$	1.1%	Not derivable in finite-symmetry algebra (Thm. A)
$\sin^2 \theta_{12} \approx 3/\pi^2$	0.003σ	Transcendental; no derivation in any tested framework
$\theta_K \approx 2/9$ (Koide)	9.5 bits	Empirical; CW dynamics fails by $178\times$
Pisano density $\approx 3/8$	6×10^{-5}	Number-theoretic conjecture; no physics derivation

These remain empirical features of SM parameter data. Theorem A establishes that within finite-symmetry algebra, no derivation is possible. Whether the patterns indicate (a) coincidences in dense parameter space, (b) signals of an infinite-dimensional structure, or (c) consequences of a yet-unidentified non-finite-symmetry organizing principle, remains open.

6. CONCLUSION

The class of finite-symmetry constructions based on the binary icosahedral group $2I$ and its standard extensions provably cannot derive the Standard Model. Three headline structural impossibility theorems close the program from independent directions. The Arithmetic Obstruction (Theorem A) bars any finite-symmetry construction from fusing icosahedral character data with the cyclotomic-14 field containing $\sin(\pi/14)$ — indeed, since $\sin(\pi/14)$ is not an algebraic integer, it can never be a character value of any finite group at all. The Frobenius–Schur chirality obstruction (Theorem B) closes all worldsheet realizations of $2I$, geometric or non-geometric: no $2I$ irrep is of complex type, every faithful one is quaternionic, and the equivariant Dirac index vanishes identically. The centralizer obstruction (Theorem C) closes heterotic bundle realization of the $F_4 \times G_2$ folded gauge group: the fold survives only as a matter-content classification, never as a dynamical 4D gauge symmetry. Five supporting propositions (Propositions 4.1–4.9) complete the closure, covering surviving-algebra structure, generation–gauge incompatibility, geometry-independence, CRT factorization of square-free modular flavor groups, and the modular-tower plateau.

The mathematical contribution is negative: a class of candidate theories ruled out by exhaustive computational and analytical analysis. The specific algebraic and structural constraints any successful future program must satisfy are made explicit. In particular, any program seeking to derive $\sin(\pi/14)$ (and hence the Cabibbo angle) from finite discrete symmetry must include the prime 7 in its symmetry’s exponent (for which $2I$ -based constructions are insufficient), or appeal to infinite-dimensional structures such as vertex operator algebras with level-mixing.

The numerical coincidences in SM parameter data motivate continued research, but the present results indicate that the path forward will require fundamentally different mathematical machinery from the finite-symmetry algebra explored here. A companion paper [20] explores the natural non-simply-laced extension via outer-automorphism folding $E_8 \supset F_4 \times G_2$ and its structural commitments.

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